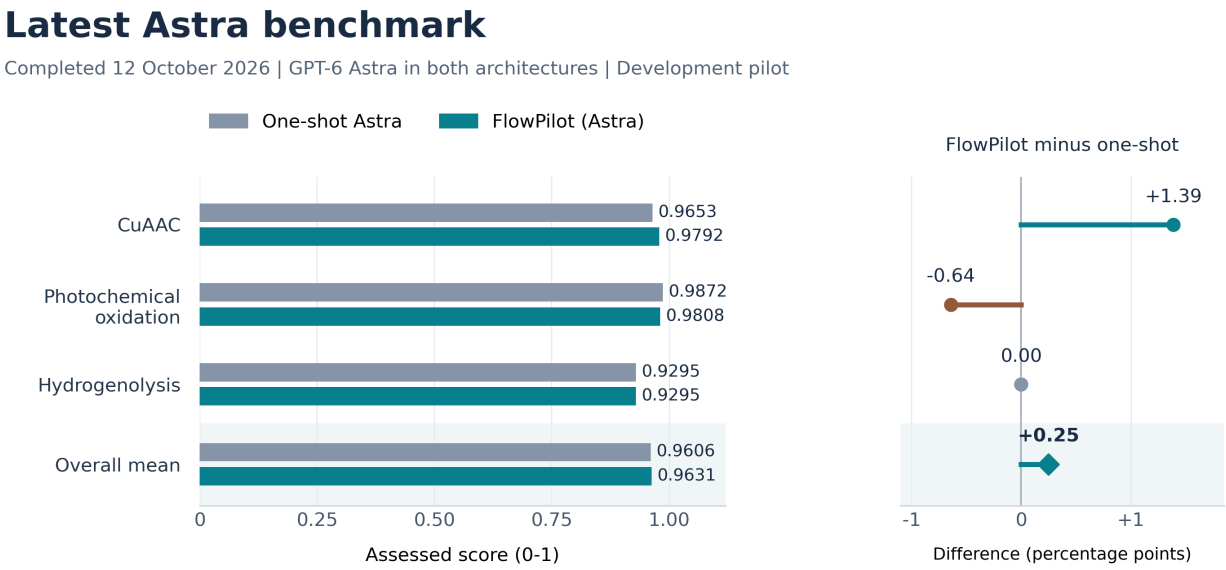
Latest Astra benchmark

Completed 12 October 2026 | Prepared 13 October 2026 | Development pilot

FlowPilot averaged 0.9631, compared with 0.9606 for one-shot Astra. The difference was +0.002493 on the 0-1 scale, or **+0.25 percentage points**. Both architectures received zero judge-reported critical flags.



How the comparison was made

GPT-6 Astra with medium reasoning generated one design per architecture for each of three chemistries (six designs). Both architectures received matched task and inventory information. FlowPilot used the cheaper legacy configuration with two initial candidates and source-excluded lexical retrieval; computation was not matched. Qwen3.6-27B, GPT-4o and Claude Sonnet 4.6 evaluated every design, giving 18 accepted judgments.

Each judge rated applicable criteria from a fixed 14-criterion rubric on a 0-4 scale. The applicable ratings were averaged and divided by four, then averaged equally across judges and chemistries. Critical flags were counted separately. Generation labels were withheld, although complete blinding was not claimed.

What the result supports

The CuAAC gain (+1.39 percentage points) outweighed the oxidation decrease (-0.64); hydrogenolysis tied. The aggregate lead is small and depends on the judge panel. Excluding GPT-4o gives **0.9607 for FlowPilot and 0.9768 for one-shot**. These cases informed development, and the two preceding completed comparisons favored one-shot. With one generation per condition, this pilot establishes a higher observed mean, but does not establish repeatability or general superiority.

Verification and recorded cost

Scores were reproduced from the accepted judge responses; all 95 recorded generation-file hashes and the matched-input audit were checked. The saved API estimate for this round is **$11.34** ($9.91 generation; $1.43 judging), excluding local hardware and unreturned billable failures. No new model calls were made to prepare this report.

Source campaign: ablation_results/manuscript_benchmark/
astra_verified_repairs_dev04_20261012
Source hashes, accepted-score reconstruction and exact values accompany the figure in verified_results.json.